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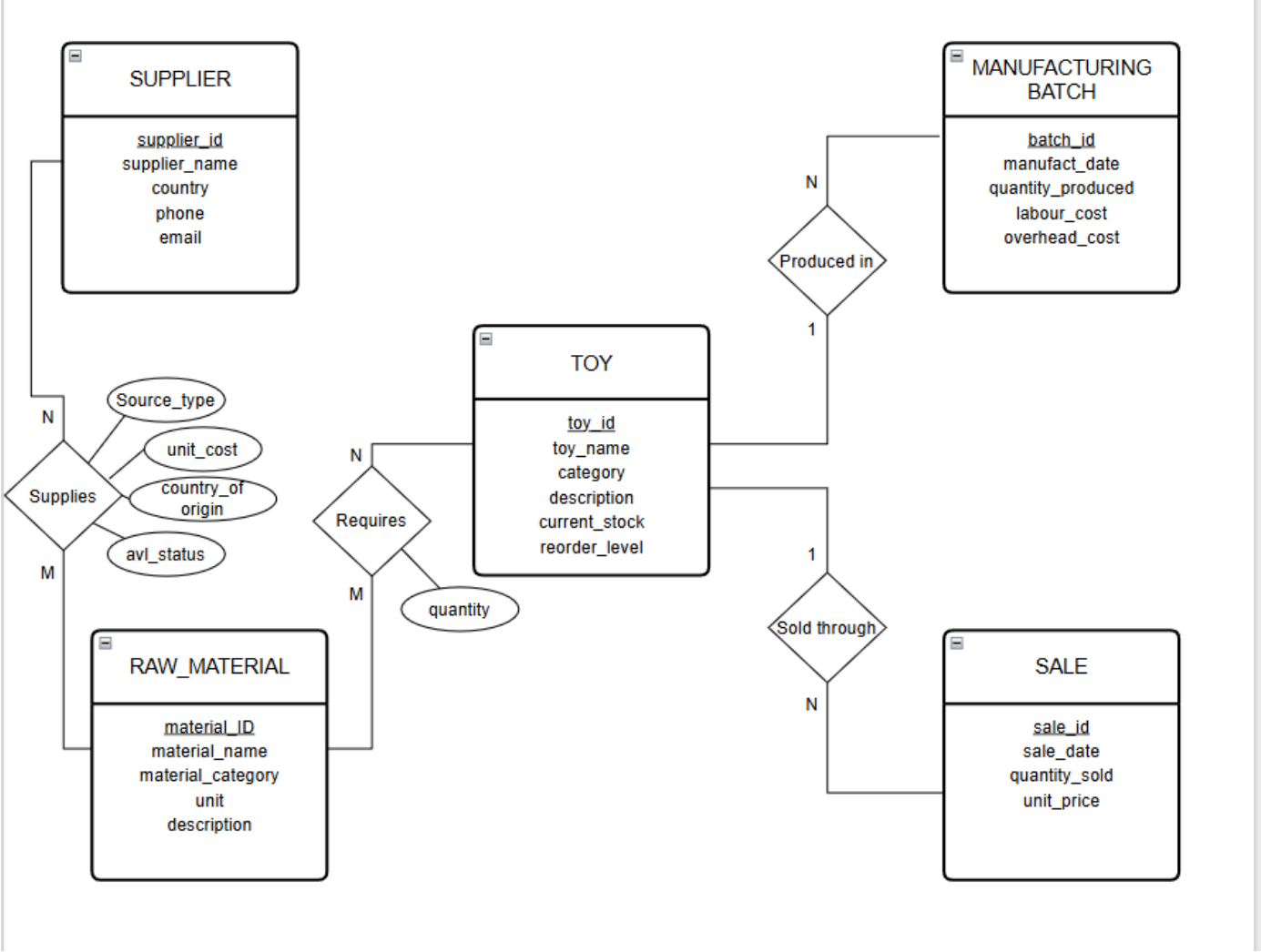
ABSTRACT

CHAPTER 1: INTRODUCTION

CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

CHAPTER 3: FUNCTIONAL REQUIREMENTS-Describe different modules.Also mention softwares being used

CHAPTER 4: ER MODEL & RELATIONAL TABLES ALONG WITH SAMPLE DATA



The conceptual ER model contains five main entities: SUPPLIER, RAW_MATERIAL, TOY, MANUFACTURING_BATCH and SALE. The M:N Supplies and Requires relationships are implemented as junction tables. Foreign keys appear in the relational schemas because they are required for implementation, although they are not drawn as attributes in the conceptual ER diagram.

Notation: underlined attributes are primary keys (PK); FK denotes a foreign key; UK denotes a unique candidate key. SUPPLIES and REQUIRES use composite primary keys.

4.1 Relational Schemas

SUPPLIER

SUPPLIER(supplier_id, supplier_name, country, phone, email)

PK: supplier_id; email may be UNIQUE.

RAW_MATERIAL

RAW_MATERIAL(material_id , material_name, material_category, unit, description)

PK: material_id; material_name may be UNIQUE.

TOY

TOY(toy_id, toy_name, category, description, current_stock, reorder_level)

PK: toy_id; stock and reorder level must be non-negative.

SUPPLIES — junction

SUPPLIES(supplier_id , material_id, source_type, unit_cost, country_of_origin, availability_status)

Composite PK: (supplier_id, material_id). Stores attributes of each supplier–material pairing.

REQUIRES — junction

REQUIRES(toy_id , material_id, quantity_required)

Composite PK: (toy_id, material_id). Stores the required quantity for each toy–material pair.

MANUFACTURING_BATCH

MANUFACTURING_BATCH(batch_id, toy_id , manufacturing_date, quantity_produced, labour_cost, overhead_cost)

PK: batch_id; toy_id comes from the 1:N Produced in relationship.

SALE

SALE(sale_id , toy_id , sale_date, quantity_sold, unit_price)

PK: sale_id; toy_id comes from the 1:N Sold through relationship.

4.2 Sample Data

The records below form a coherent illustrative dataset. Monetary amounts are in Indian rupees (₹); dates are shown as YYYY-MM-DD.

Table: SUPPLIER

supplier_id	supplier_name	country	phone	email
1	Coastal Polymer Supplies	India	9876501001	sales@coastalpolymer.example
2	East Asia Components	China	861380010002	sales@eastasia.example
3	Western Metal Traders	India	9876501003	orders@westernmetal.example
4	SafeColor Industries	India	9876501004	support@safecolor.example
5	Bengal Textile Supply	Bangladesh	880170010005	sales@bengaltextile.example

Table: RAW_MATERIAL

material_id	material_name	material_category	unit	description
101	Plastic Granules	Polymer	kg	For molded toy parts
102	Steel Wire	Metal	kg	For axles and frames
103	Non-Toxic Paint	Finishing	litre	Child-safe finish
104	Cotton Fabric	Textile	metre	For soft toys
105	Electronic Module	Electronic	piece	For electronic toys

Table: TOY

toy_id	toy_name	category	description	current_stock	reorder_level
201	Building Blocks	Construction	Interlocking plastic blocks	90	20
202	Mini Racer Car	Vehicle	Small toy car with steel axle	68	15
203	Cuddle Doll	Doll	Fabric doll with painted details	55	12
204	Smart Robot	Electronic	Battery-powered toy robot	32	10
205	Plush Bear	Soft Toy	Stuffed teddy bear	61	15

Table: SUPPLIES

supplier_id	material_id	source_type	unit_cost	country_of_origin	availability_status
1	101	LOCAL	120.00	India	AVAILABLE
2	101	IMPORTED	105.00	China	AVAILABLE
3	102	LOCAL	80.00	India	AVAILABLE
4	103	LOCAL	150.00	India	AVAILABLE
5	104	IMPORTED	90.00	Bangladesh	LIMITED
1	104	LOCAL	98.00	India	AVAILABLE
2	105	IMPORTED	250.00	China	AVAILABLE

Table: REQUIRES

toy_id	material_id	quantity_required
201	101	0.200 kg
201	103	0.050 litre
202	101	0.150 kg
202	102	0.080 kg
203	104	0.600 metre
203	103	0.030 litre
204	105	1 piece
204	101	0.100 kg
205	104	0.500 metre
205	103	0.020 litre

Table: MANUFACTURING_BATCH

batch_id	toy_id	manufacturing_date	quantity_produced	labour_cost	overhead_cost
301	201	2026-08-01	100	4500.00	1800.00
302	202	2026-08-05	80	3200.00	1200.00
303	203	2026-08-12	60	3900.00	1500.00
304	204	2026-08-20	40	5200.00	2200.00
305	205	2026-08-25	70	2800.00	1100.00

Table: SALE

sale_id	toy_id	sale_date	quantity_sold	unit_price
401	201	2026-09-02	10	499.00
402	202	2026-09-04	12	349.00
403	203	2026-09-06	5	799.00
404	204	2026-09-10	8	1499.00
405	205	2026-09-12	9	599.00

4.3 Validation Notes

- All five main entities and both junction tables are represented.
- Primary keys are unique; composite key pairs in SUPPLIES and REQUIRES are not repeated.
- Foreign keys refer to existing suppliers, materials, toys, batches, and sales parents as appropriate.

- SUPPLIES includes both LOCAL and IMPORTED sourcing. Plastic Granules and Cotton Fabric each have two sourcing options.
- Stock values are illustrative snapshots and are not automatically reconciled with sample production and sale records.

CHAPTER 5: DDL COMMANDS & PL/SQL PROCEDURES / FUNCTIONS / TRIGGERS

The following Oracle examples implement the seven-table design. Create parent tables before dependent tables.

```
CREATE TABLE SUPPLIER (
 supplier_id NUMBER(10) PRIMARY KEY,
 supplier_name VARCHAR2(100) NOT NULL,
 country VARCHAR2(60) NOT NULL,
 phone VARCHAR2(20),
 email VARCHAR2(120) UNIQUE
);

CREATE TABLE RAW_MATERIAL (
 material_id NUMBER(10) PRIMARY KEY,
 material_name VARCHAR2(100) NOT NULL UNIQUE,
 material_category VARCHAR2(50) NOT NULL,
 unit VARCHAR2(20) NOT NULL,
 description VARCHAR2(300)
);

CREATE TABLE TOY (
 toy_id NUMBER(10) PRIMARY KEY,
 toy_name VARCHAR2(100) NOT NULL,
 category VARCHAR2(50) NOT NULL,
 description VARCHAR2(300),
 current_stock NUMBER(10) DEFAULT 0 NOT NULL CHECK (current_stock >= 0),
 reorder_level NUMBER(10) DEFAULT 0 NOT NULL CHECK (reorder_level >= 0)
);

CREATE TABLE SUPPLIES (
 supplier_id NUMBER(10) NOT NULL REFERENCES SUPPLIER(supplier_id),
 material_id NUMBER(10) NOT NULL REFERENCES RAW_MATERIAL(material_id),
 source_type VARCHAR2(10) NOT NULL CHECK (source_type IN ('LOCAL','IMPORTED')),
 unit_cost NUMBER(10,2) NOT NULL CHECK (unit_cost >= 0),
 country_of_origin VARCHAR2(60) NOT NULL,
 availability_status VARCHAR2(20) NOT NULL
   CHECK (availability_status IN ('AVAILABLE','LIMITED','UNAVAILABLE')),
 CONSTRAINT pk_supplies PRIMARY KEY (supplier_id, material_id)
);

CREATE TABLE REQUIRES (
 toy_id NUMBER(10) NOT NULL REFERENCES TOY(toy_id),
 material_id NUMBER(10) NOT NULL REFERENCES RAW_MATERIAL(material_id),
 quantity_required NUMBER(12,3) NOT NULL CHECK (quantity_required > 0),
 CONSTRAINT pk_requires PRIMARY KEY (toy_id, material_id)
);

CREATE TABLE MANUFACTURING_BATCH (
 batch_id NUMBER(10) PRIMARY KEY,
 toy_id NUMBER(10) NOT NULL REFERENCES TOY(toy_id),
 manufacturing_date DATE NOT NULL,
 quantity_produced NUMBER(10) NOT NULL CHECK (quantity_produced > 0),
 labour_cost NUMBER(12,2) DEFAULT 0 NOT NULL CHECK (labour_cost >= 0),
 overhead_cost NUMBER(12,2) DEFAULT 0 NOT NULL CHECK (overhead_cost >= 0)
);

CREATE TABLE SALE (
 sale_id NUMBER(10) PRIMARY KEY,
```

```
 toy_id NUMBER(10) NOT NULL REFERENCES TOY(toy_id),
 sale_date DATE NOT NULL,
 quantity_sold NUMBER(10) NOT NULL CHECK (quantity_sold > 0),
 unit_price NUMBER(10,2) NOT NULL CHECK (unit_price >= 0)
);
```

5.2 Function: Calculate Gross Sale Amount

Returns quantity_sold multiplied by unit_price for a specified sale.

```
CREATE OR REPLACE FUNCTION sale_amount(p_sale_id IN NUMBER)
RETURN NUMBER
IS
 v_amount NUMBER(12,2);
BEGIN
 SELECT quantity_sold * unit_price
 INTO v_amount
 FROM SALE
 WHERE sale_id = p_sale_id;
 RETURN v_amount;
EXCEPTION
 WHEN NO_DATA_FOUND THEN RETURN NULL;
END;
/
```

5.3 Procedure: Record a Sale and Update Stock

Checks available stock, inserts the sale and decreases current_stock. The caller can COMMIT or ROLLBACK the transaction.

```
CREATE OR REPLACE PROCEDURE record_sale (
 p_sale_id IN NUMBER, p_toy_id IN NUMBER, p_sale_date IN DATE,
 p_quantity IN NUMBER, p_unit_price IN NUMBER
)
IS
 v_stock TOY.current_stock%TYPE;
BEGIN
 SELECT current_stock INTO v_stock
 FROM TOY WHERE toy_id = p_toy_id FOR UPDATE;

 IF p_quantity <= 0 OR p_quantity > v_stock THEN
   RAISE_APPLICATION_ERROR(-20001, 'Invalid quantity or insufficient stock');
 END IF;

 INSERT INTO SALE(sale_id, toy_id, sale_date, quantity_sold, unit_price)
 VALUES (p_sale_id, p_toy_id, p_sale_date, p_quantity, p_unit_price);

 UPDATE TOY SET current_stock = current_stock - p_quantity
 WHERE toy_id = p_toy_id;
END;
/
```

5.4 Trigger: Add Produced Units to Stock

Adds the quantity produced to the corresponding toy's current stock after a new batch is inserted.

```
CREATE OR REPLACE TRIGGER trg_batch_add_stock
AFTER INSERT ON MANUFACTURING_BATCH
FOR EACH ROW
BEGIN
 UPDATE TOY
 SET current_stock = current_stock + :NEW.quantity_produced
 WHERE toy_id = :NEW.toy_id;
END;
/
```

Note: if this trigger is used, do not manually add the same batch quantity to stock again. Extra logic is needed if batches are later updated or deleted.

5.5 Useful Queries

```
-- Toys at or below the reorder level
SELECT toy_id, toy_name, current_stock, reorder_level
FROM TOY
WHERE current_stock <= reorder_level;

-- Compare local and imported sourcing options
SELECT r.material_name, s.source_type, s.country_of_origin,
       s.unit_cost, s.availability_status
FROM RAW_MATERIAL r JOIN SUPPLIES s ON s.material_id = r.material_id
ORDER BY r.material_name, s.unit_cost;

-- Gross revenue by toy
SELECT t.toy_name, SUM(s.quantity_sold * s.unit_price) AS gross_revenue
FROM TOY t JOIN SALE s ON s.toy_id = t.toy_id
GROUP BY t.toy_name;
```

CHAPTER 6: RESULTS & SNAPSHOTS

The design supports supplier and raw-material sourcing management, toy material requirements, manufacturing batch records, current-stock tracking and toy-wise sales analysis. After running the DDL and inserting sample data in Oracle Database 21c XE, capture screenshots of the following results.

Test	Expected result
SELECT from TOY and RAW_MATERIAL	Displays five toy records and five raw-material records.
Sourcing comparison query	Shows LOCAL and IMPORTED options, origin, unit cost and availability.
BOM query	Shows the raw materials and quantities required by each toy.
Manufacturing batch query	Displays batch date, produced quantity, labour cost and overhead.
sale_amount function	Returns quantity_sold × unit_price for a valid sale ID.
Reorder query	Lists toys where current_stock <= reorder_level.
record_sale procedure	Inserts a sale and decreases stock if sufficient stock is available.
Batch trigger	Increases stock when a manufacturing batch is inserted.

6.1 Screenshots

Insert actual screenshots from your own Oracle execution. Do not describe expected output as an executed result until you have run the statements.

CHAPTER 7: CONCLUSION, LIMITATIONS & FUTURE WORK

7.1 Conclusion

The Toy Manufacturing and Import Dependency Analysis System organizes supplier, raw-material, toy, manufacturing and sales information in a compact relational design. Supplies records sourcing type, origin, cost and availability, while Requires records the quantity of each material needed per toy. The database supports comparisons of local and imported sourcing options, basic stock management, manufacturing records and gross sales analysis. Oracle constraints and PL/SQL routines help maintain data integrity and automate selected operations.

7.2 Limitations

- The model does not record actual material consumption for each manufacturing batch.
- Sales are linked directly to toys, so individual sales cannot be traced to a specific production batch.
- Inventory is stored as a current-stock value; stock movement history and multiple warehouses are not represented.
- Sales channels, discounts and platform fees are not separately modeled.
- The sourcing comparison does not automatically determine whether different materials are technically interchangeable.

7.3 Future Work

- Add batch-level material usage to record actual quantities and costs consumed during production.
- Add sales channels, discounts and platform fees for more detailed net-revenue analysis.
- Maintain inventory transaction history and automated low-stock alerts.
- Generate reports comparing estimated production costs under local and imported sourcing scenarios.
- Add supplier lead times and validated material-substitution rules.

CHAPTER 8: REFERENCES

The following standard references cover database design and the Oracle technologies used in the project.

1. Oracle. Oracle Database SQL Language Reference, 21c. Oracle Documentation.
2. Oracle. Oracle Database PL/SQL Language Reference, 21c. Oracle Documentation.
3. Elmasri, R. and Navathe, S. B. Fundamentals of Database Systems. Pearson.
4. Silberschatz, A., Korth, H. F., and Sudarshan, S. Database System Concepts. McGraw Hill.

Add any course notes or specific online resources actually consulted, using the reference format required by your institution.